

M. WAHAJ NAVEED

Software Engineer | Full Stack & ML Learner

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I'm a dedicated Software Engineering graduate from COMSATS University Islamabad with solid skills in full-stack development and adding ML to apps. In group projects, I've helped build scalable web systems and microservices using Python, React, Node.js, and Docker—cutting processing time by 40% through teamwork. I love collaborating with others to turn ideas into reliable solutions. As a Software Engineer, I'm excited to join a team, share my skills, learn from everyone, and help create great stuff together.

Skills

- React.js
- Node.js
- Python
- TypeScript
- C++
- SQL
- Docker
- Git/GitHub
- Next.js
- Flask
- PostgreSQL
- MongoDB
- Scikit-learn
- Figma
- Linux

Education

BS Software Engineering

COMSATS University Islamabad

2022 – 2026

Relevant Coursework

- CGPA: 3.59 / 4.00
- Honors: PM Laptop Award Recipient

- Data Structures & Algorithms
- Design & Architecture
- Information Security
- Requirements Engineering

Experience

Frontend Engineer Trainee

CapregSoft (Pvt) Ltd

Aug 2024 – March 2025

Wah, Pakistan

React.js, Docker, Github

- **Project FusionIQX:** Engineered the frontend architecture for an industry-grade analytical dashboard using React.js, optimizing data visualization performance.
- **Airport Taxihub:** Developed the core booking interface with strict security protocols.
- Led the integration of Docker containerization ensuring consistent dev environments.

Front-End Developer Trainee

Meta (Coursera)

April 2024

Remote

React.js, UI/UX

- Developed "Little Lemon" functional web app, achieving a 98/100 Lighthouse performance score.
- Mastered advanced React hooks for seamless mobile/desktop experiences.

Key Projects

ClarityWorks (Final Year Project)

AI-Powered Readability Analysis Platform

In Progress

Full Stack + ML

Microservices, Python, React, NLP, PostgreSQL

- **System Architecture:** Designed a robust Three-Tier Architecture utilizing a decoupled Microservices pattern. The system integrates a React/TypeScript frontend, a Node.js business logic layer, and a dedicated Python/Flask service for ML computation.
- **ML Pipeline & Ensemble Learning:** Developed an advanced prediction engine using **Random Forest** and **Gradient Boosting Regressors**. Trained on the **CLEAR Corpus**, the model achieves high-confidence classification of text complexity across K-12 and collegiate grade levels.
- **Hybrid Analysis Engine:** Combined classical readability formulas (Flesch-Kincaid, SMOG, ARI) with modern linguistic feature extraction. Implemented **Dale-Chall** algorithms for difficult word detection and syllable counting logic using pyphen.

- **Feature Engineering:** Engineered extraction pipelines for 11 key linguistic features including lexical diversity (Type-Token Ratio), syntactic complexity, and sentence entropy to drive the predictive models.
- **Security & Data Flow:** Implemented secure **JWT Authentication** with bcrypt password hashing. Designed a normalized **PostgreSQL** schema using JSONB for efficient storage of complex analysis reports.

JAAW - Video Conferencing Platform

Real-time Communication System

React, TypeScript, Redux, Firebase, Zego Cloud

2024
Academic

- Built a real-time web video conferencing website with robust multi-user support, enabling seamless group communication.
- Integrated advanced collaboration features including screen sharing, secure call recording, and real-time chat functionality.
- Implemented state management using Redux and utilized Firebase for backend services to ensure scalable data handling.

Airbnb Clone (E-Commerce)

UI/UX Prototype & Implementation

Figma, HCI Principles

2024
Academic

- Designed high-fidelity prototypes following HCI principles for optimal user navigation.
- Implemented responsive layouts and user journey flows focusing on accessibility.

Algorithmic Racing Simulation

Real-time Pathfinding

C++, Data Structures

2024
Academic

- Implemented complex maze-solving algorithms using Graph Theory.
- Optimized pathfinding efficiency for real-time simulation performance.

Leadership & Awards

Leadership

- **Software Engineering Head (COMSATS Sports Society):** Deployed custom portal for society operations & event registrations.
- Spearheaded technical workshops for junior students.

Awards

- **PM Laptop Scheme:** Awarded for outstanding academic merit.